WEB HOSTING IN EC2 USING LINUX

Web hosting in Amazon Web Services EC2 involves launching a Linux instance and hosting a website using a web server. In this setup, an EC2 instance acts as a virtual server where web files are stored and served to users through the internet.

Web Server

A web server is software that stores, processes, and delivers web pages to users over the internet using HTTP or HTTPS. In Amazon Web Services EC2, a web server runs on a Linux instance to host and serve websites.

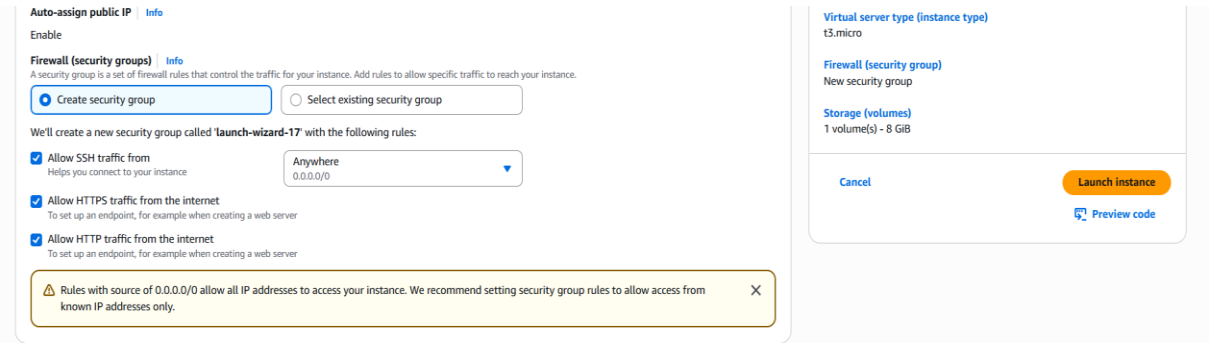
Common Web Servers

- Apache (httpd)
- Nginx
- Lighttpd

HANDS-ON FOR WEBHOSTING IN HTTPD

Screenshot 1: Launching Linux EC2 Instance

- Linux EC2 instance is launched in **Amazon Web Services**
- HTTP traffic is enabled
- HTTPS traffic is enabled
- Instance is created successfully



Screenshot: Installing Apache Web Server on Linux EC2

- Switched to root user.

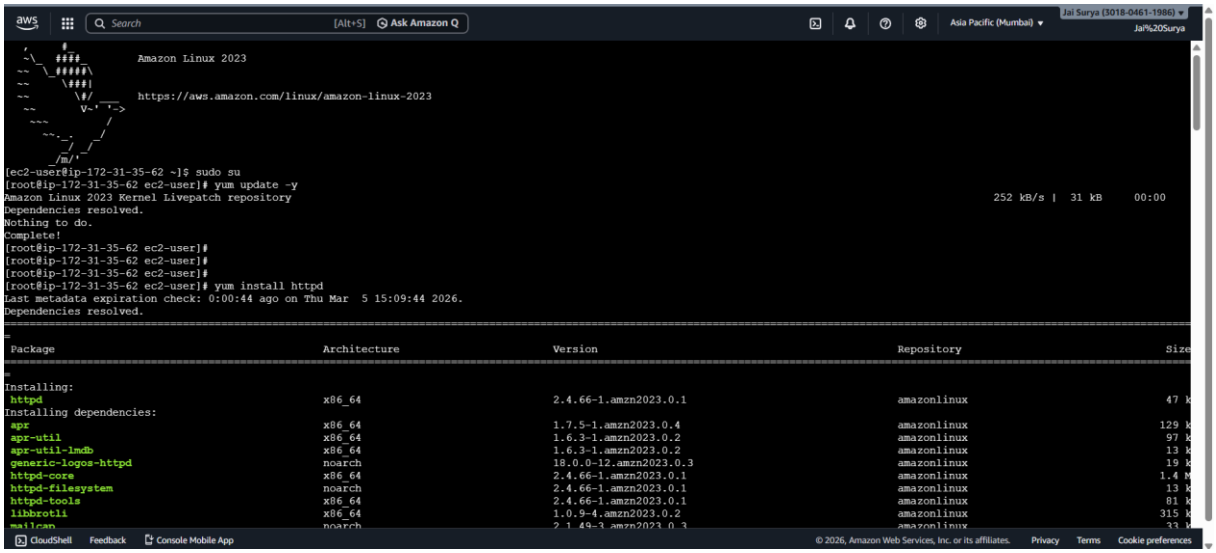
  sudo su

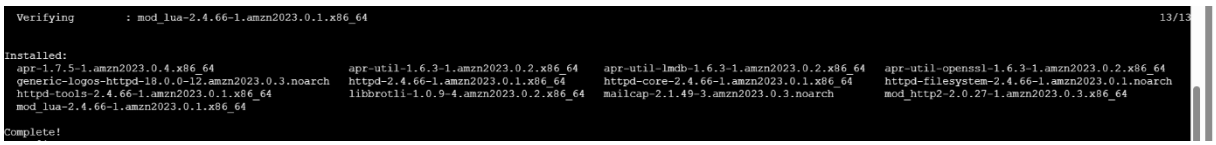
- Updated system packages.

  yum update -y

- Installed Apache web server.

  yum install httpd -y





Screenshot: Managing Apache Web Server Service

- Checked the status of the Apache service.

  systemctl status httpd

- Started the Apache web server.

  systemctl start httpd

- Verified the Apache service status again.

  systemctl status httpd

```
[root@ip-172-31-35-62 ec2-user]#
[root@ip-172-31-35-62 ec2-user]# systemctl status httpd
○ httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: inactive (dead)
       Docs: man:httpd.service(8)
[root@ip-172-31-35-62 ec2-user]#
[root@ip-172-31-35-62 ec2-user]# systemctl start httpd
[root@ip-172-31-35-62 ec2-user]# systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
     Active: active (running) since Thu 2026-03-05 15:11:38 UTC; 10s ago
       Docs: man:httpd.service(8)
   Main PID: 25732 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1067)
     Memory: 13.3M
        CPU: 75ms
     CGroup: /system.slice/httpd.service
             ├─25732 /usr/sbin/httpd -DFOREGROUND
             ├─25733 /usr/sbin/httpd -DFOREGROUND
             ├─25734 /usr/sbin/httpd -DFOREGROUND
             ├─25735 /usr/sbin/httpd -DFOREGROUND
             └─25761 /usr/sbin/httpd -DFOREGROUND

Mar 05 15:11:38 ip-172-31-35-62.ap-south-1.compute.internal systemd[1]: Starting httpd.service - The Apache HTTP Server...
Mar 05 15:11:38 ip-172-31-35-62.ap-south-1.compute.internal systemd[1]: Started httpd.service - The Apache HTTP Server.
Mar 05 15:11:38 ip-172-31-35-62.ap-south-1.compute.internal httpd[25732]: Server configured, listening on: port 80
```

Screenshot: Downloading Content Using wget

- Downloaded web content using wget with server response headers and content disposition option.

  wget -S --content-disposition <download-link>

```
[root@ip-172-31-35-62 ec2-user]# wget -S --content-disposition https://templatemo.com/download/templatemo_604_christmas_piano
--2026-03-05 15:18:11--  https://templatemo.com/download/templatemo_604_christmas_piano
Resolving templatemo.com (templatemo.com)... 72.52.176.250
Connecting to templatemo.com (templatemo.com)|72.52.176.250|:443... connected.
HTTP request sent, awaiting response...
```

Screenshot: Extracting ZIP File

- Extracted the downloaded ZIP file using the unzip command.

  unzip <filename>

```
2026-03-05 15:18:17 (222 KB/s) - 'templatemo_604_christmas_piano.zip' saved [649787/649787]

[root@ip-172-31-35-62 ec2-user]# ls
templatemo_604_christmas_piano.zip
[root@ip-172-31-35-62 ec2-user]# unzip templatemo_604_christmas_piano.zip
Archive:  templatemo_604_christmas_piano.zip
   creating: templatemo_604_christmas_piano/
   creating: templatemo_604_christmas_piano/images/
  inflating: templatemo_604_christmas_piano/images/about-piano.jpg
  inflating: templatemo_604_christmas_piano/images/christmas-pic-01.jpg
  inflating: templatemo_604_christmas_piano/images/christmas-pic-02.jpg
  inflating: templatemo_604_christmas_piano/images/christmas-pic-03.jpg
  inflating: templatemo_604_christmas_piano/images/christmas-pic-04.jpg
  inflating: templatemo_604_christmas_piano/images/christmas-pic-05.jpg
  inflating: templatemo_604_christmas_piano/images/christmas-pic-06.jpg
  inflating: templatemo_604_christmas_piano/images/christmas-pic-07.jpg
  inflating: templatemo_604_christmas_piano/images/christmas-pic-08.jpg
  inflating: templatemo_604_christmas_piano/index.html
  inflating: templatemo_604_christmas_piano/templatemo-604-christmas-piano.css
  inflating: templatemo_604_christmas_piano/templatemo-604-piano-scripts.js
[root@ip-172-31-35-62 ec2-user]# ls
templatemo_604_christmas_piano  templatemo_604_christmas_piano.zip
[root@ip-172-31-35-62 ec2-user]#
```

Screenshot: Moving Website Files to Web Root Directory

- Moved into the extracted project folder.

  cd <folder-name>

- Moved all website files to the Apache web root directory.

  mv * /var/www/html/

- Listed the files to verify the content.

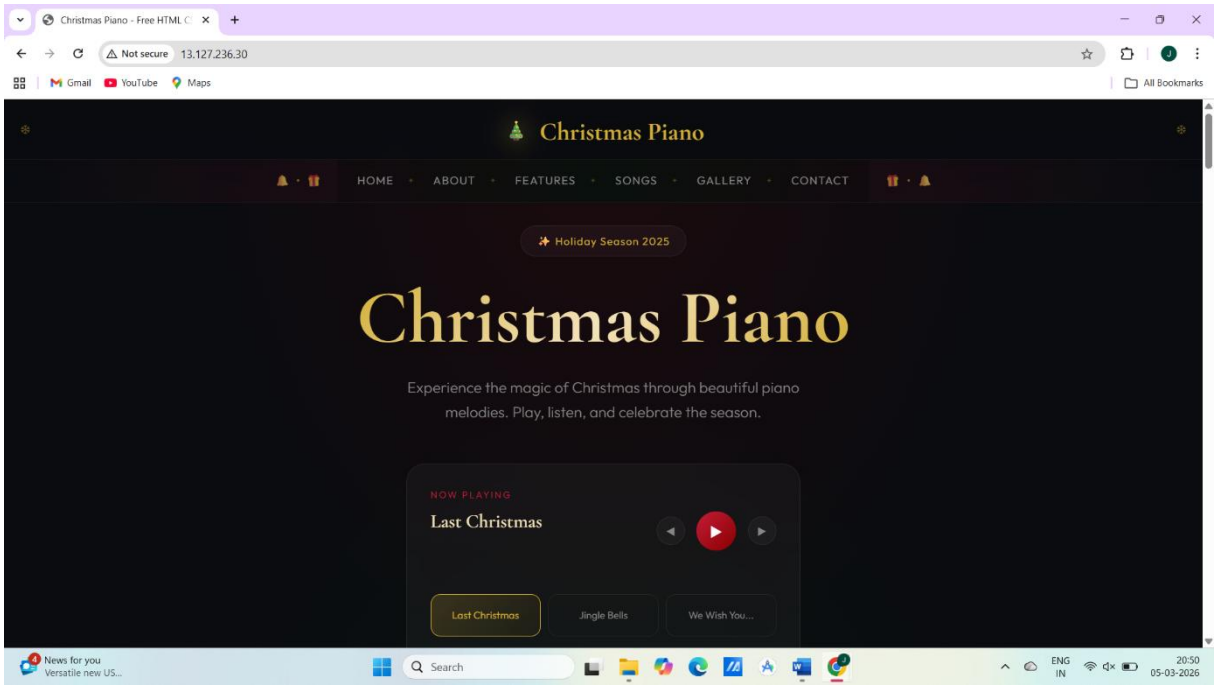
  ls

```
[root@ip-172-31-35-62 ec2-user]#
[root@ip-172-31-35-62 ec2-user]#
[root@ip-172-31-35-62 ec2-user]# cd templatemo_604_christmas_piano/
[root@ip-172-31-35-62 templatemo_604_christmas_piano]# ls
images  index.html  templatemo-604-christmas-piano.css  templatemo-604-piano-scripts.js
[root@ip-172-31-35-62 templatemo_604_christmas_piano]# mv * /var/www/html/
[root@ip-172-31-35-62 templatemo_604_christmas_piano]# ls
[root@ip-172-31-35-62 templatemo_604_christmas_piano]# cd /var/www/html/
[root@ip-172-31-35-62 html]# ls
images  index.html  templatemo-604-christmas-piano.css  templatemo-604-piano-scripts.js
[root@ip-172-31-35-62 html]#
[root@ip-172-31-35-62 html]#
```

Screenshot: Accessing Website Using Public IP

- Copied the public IP address of the EC2 instance.
- Pasted the public IP into a new browser tab.

  http://<public-ip>



HOSTING WEBSITE USING NGINX WEB SERVER

Screenshot: Installing and Hosting Website Using Nginx Web Server

- Installed the Nginx web server on the Linux EC2 instance.

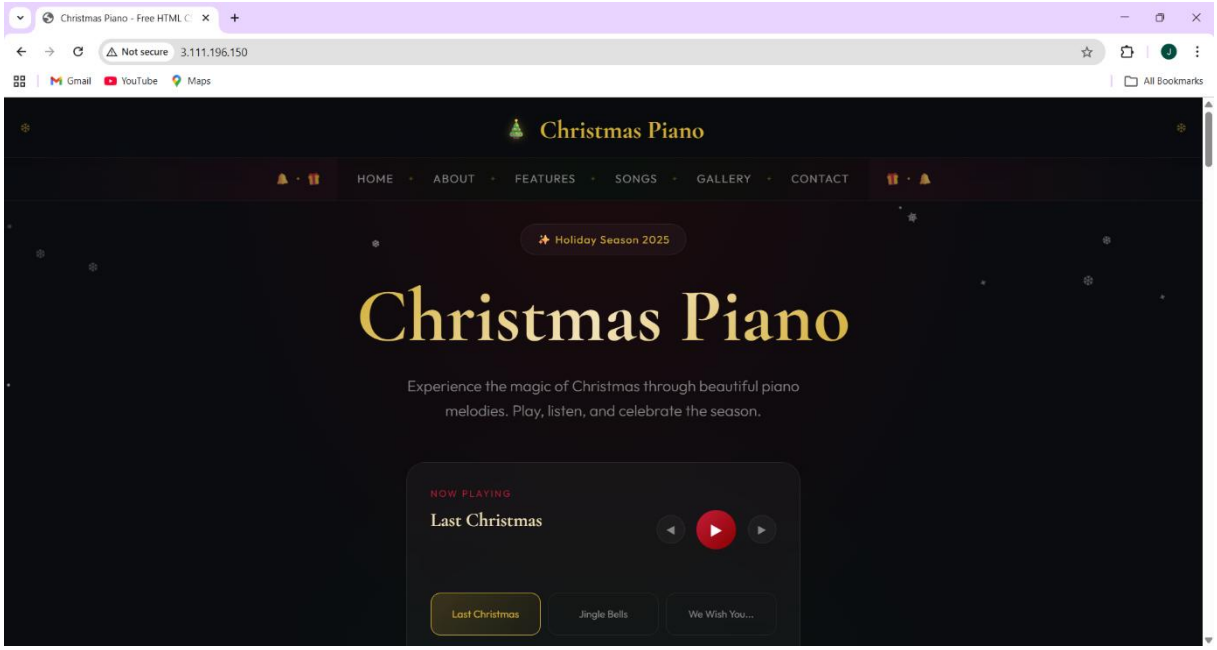
  yum install nginx -y
- Started the Nginx service.

  systemctl start nginx
- Checked the status of the Nginx service.

  systemctl status nginx
- Moved website files to the Nginx web root directory.

  mv * /usr/share/nginx/html/

- Accessed the website using the EC2 public IP in a browser.

  http://<public-ip>

```
[root@ip-172-31-41-143 html]# history
    1  yum update -y
    2  yum install nginx
    3  systemctl status nginx
    4  systemctl start nginx
    5  systemctl status nginx
    6  wget -S --content-disposition https://templatemo.com/download/templatemo_604_christmas_piano
    7  ls
    8  unzip templatemo_604_christmas_piano.zip
    9  ls
   10  cd templatemo_604_christmas_piano/
   11  ls
   12  mv * /usr/share/nginx/html/
   13  ls
   14  cd /usr/share/nginx/html/
   15  ls
   16  history
[root@ip-172-31-41-143 html]#
```